# Edhec Business School



# SOLUTIONS: HW #2

**Q1. Derivative pricing and hedging**

In this question, we study the prices of bonds of different maturities in the Vasicek (1977) model. We will assume that there is a single source of uncertainty. Let this "single factor" be the yield $r(t)$ on the instant-maturity bond.

Denote by $t$ the date today, by $T_k$ the maturity date of the $k^{\text{th}}$ bond, and by $\tau_k = T_k - t$ the time-to-maturity of this bond. Denote by $P(t, \tau_k)$ the price of a discount bond that makes a single payment of \$1 in $\tau_k$ periods from today. Thus, the price of a discount bond with 0 time to maturity is: $P(t, 0) = 1$, and the instantaneous rate of return on a bond is given by: $\frac{dP(t,\tau_k)}{P(t,\tau_k)}$. The instantaneous yield, $r(t)$, then is defined as

$$\lim_{\tau \to 0} \frac{dP(t,\tau)}{P(t,\tau)} = r(t)\, dt.$$

The Vasicek model assumes that $r(t)$ follows an Ornstein-Uhlenbeck process:

$$dr(t) = \kappa\big[r_\infty - r(t)\big]\, dt + \sigma_r\, dB_r,$$

where $\kappa, r_\infty$ and $\sigma_r$ are positive constants, and $B_r$ is a Brownian motion under the physical measure.

Please answer the following questions.

**1.1** Assume that prices for bonds of all maturities depend on only a single factor, $r(t)$. Denote the price of a $\tau$-maturity bond by $P[r(t), \tau_k]$. Then, write down the process for the return on this bond, that is, $\frac{dP[r(t),\tau_k]}{P[r(t),\tau_k]}$; that is, find the mean and volatility of the bond's rate of return, denoted by $\mu_p(r, \tau_k)$ and $\sigma_p(\tau_k)$, respectively. *Note that this question does not require you to solve for $P[r(t), \tau_k]$ in terms of the interest rate; your answer should be in terms of $P[r(t), \tau_k]$ and its derivatives.*

**1.2** The Sharpe ratio (or, equivalently, the market price of risk) is the expected return in excess of the risk-free rate divided by the volatility of the return. Assume that the Sharpe ratio of a bond is constant over time; denote this constant by $SR$. Use the expressions for the mean and volatility of the bond's rate of return to derive the differential equation in terms of $SR$ that bonds of all maturities must satisfy in this model.

**1.3** Consider a portfolio $H(t)$ that consists of two assets: 1 unit long of a bond of maturity $\tau_1$ and $x$ units short of a bond with maturity $\tau_2$. Write down the rate of return on this portfolio. Find the expression for $x$ so that the rate of the return on portfolio $H(t)$ is risk-free.

**1.4** Use the expression for the rate of return on portfolio $H(t)$ to derive an expression relating the Sharpe ratio of the first bond with maturity $\tau_1$ to the Sharpe ratio of the second bond with maturity $\tau_2$. Provide an economic interpretation for this condition.

**1.1** *Assume that prices for bonds of all maturities depend on only a single factor, $r(t)$. Denote the price of a $\tau$-maturity bond by $P[r(t), \tau_k]$. Then, write down the process for the return on this bond, that is, $\frac{dP[r(t),\tau_k]}{P[r(t),\tau_k]}$; that is, find the mean and volatility of the bond's rate of return, denoted by $\mu_p(r, \tau_k)$ and $\sigma_p(\tau_k)$, respectively.*

**Solution:**
Applying Itô's formula to $P(r, \tau_k)$, where $\tau_k = T_k - t$, we get:

$$\begin{aligned} dP(r,\tau_k) &= P_t\,dt + P_r\,dr + \frac{1}{2}P_{rr}(dr)^2 \\ &= P_t\,dt + P_r\,\big(\kappa\big[r_\infty - r(t)\big]\,dt + \sigma_r\,dB_r\big) + \frac{1}{2}P_{rr}\sigma_r^2\,dt \\ &= \Big(P_t + P_r\,\kappa\big[r_\infty - r(t)\big] + \frac{1}{2}P_{rr}\sigma_r^2\Big)dt + P_r\,\sigma_r\,dB_r \end{aligned}$$

so that the return on the bond is given by:

$$\begin{aligned} \frac{dP(r,\tau_k)}{P(r,\tau_k)} &= \frac{1}{P(r,\tau_k)}\Big(P_t + P_r\,\kappa\big[r_\infty - r(t)\big] + \frac{1}{2}P_{rr}\sigma_r^2\Big)dt + \frac{1}{P(r,\tau_k)}P_r\,\sigma_r\,dB_r \\ &= \mu_p(r,\tau_k)\,dt + \sigma_p(r,\tau_k)\,dB_r, \end{aligned}$$

where

$$\begin{aligned} \mu_p(r,\tau_k) &= \frac{1}{P(r,\tau_k)}\Big(P_t + P_r\,\kappa\big[r_\infty - r(t)\big] + \frac{1}{2}P_{rr}\sigma_r^2\Big) \\ \sigma_p(r,\tau_k) &= \frac{1}{P(r,\tau_k)}P_r\,\sigma_r. \end{aligned}$$

The above can also be written as:

$$dP(r,\tau_k) = \mu_p(r,\tau_k)P(r,\tau_k)\,dt + \sigma_p(r,\tau_k)\,P(r,\tau_k)dB_r.$$

**1.2** *The Sharpe ratio (or, equivalently, the market price of risk) is the expected return in excess of the risk-free rate divided by the volatility of the return. Assume that the Sharpe ratio of a bond is constant over time; denote this constant by $SR$. Use the expressions for the mean and volatility of the bond's rate of return to derive the differential equation in terms of $SR$ that bonds of all maturities must satisfy in this model.*

**Solution:**
We start by writing the definition of the Sharpe ratio for a bond with time-to-maturity $\tau_k$:

$$\begin{aligned} SR &= \frac{\mu_p(r,\tau_k) - r(t)}{\sigma_p(r,\tau_k)} \\ SR\,\sigma_p(r,\tau_k) &= \mu_p(r,\tau_k) - r(t) \\ r(t) &= \mu_p(r,\tau_k) - SR\sigma_p(r,\tau_k) \\ r(t)\,P &= P_t + P_r\,\kappa\big[r_\infty - r(t)\big] + \frac{1}{2}P_{rr}\sigma_r^2 - SR\,P_r\,\sigma_r \\ r(t)\,P &= P_t + P_r\,\big(\kappa\big[r_\infty - r(t)\big] - SR\,\sigma_r\big) + \frac{1}{2}P_{rr}\sigma_r^2, \end{aligned}$$

which is the differential equation that the bond price must satisfy.

**1.3** *Consider a portfolio $H(t)$ that consists of two assets: 1 unit long of a bond of maturity $\tau_1$ and $x$ units short of a bond with maturity $\tau_2$. Write down the rate of return on this portfolio. Find the expression for $x$ so that the rate of the return on portfolio $H(t)$ is risk-free.*

**Solution:**
We are given that

$$H(t) = 1\,P(r,\tau_1) - x\,P(r,\tau_2).$$

Thus,

$$\begin{aligned} dH &= 1\,dP(r,\tau_1) - x\,dP(r,\tau_2) \\ &= \Big[\mu_p(r,\tau_1)P(r,\tau_1)\,dt + \sigma_p(r,\tau_1)\,P(r,\tau_1)dB_r\Big] \\ &\quad - x\Big[\mu_p(r,\tau_2)P(r,\tau_2)\,dt + \sigma_p(r,\tau_2)\,P(r,\tau_2)dB_r\Big] \end{aligned}$$

which, upon collecting the deterministic and stochastic terms, gives:

$$\begin{aligned} &= \Big[\mu_p(r,\tau_1)P(r,\tau_1) - x\mu_p(r,\tau_2)P(r,\tau_2)\Big]dt \\ &\quad + \Big[\sigma_p(r,\tau_1)\,P(r,\tau_1) - x\sigma_p(r,\tau_2)\,P(r,\tau_2)\Big]dB_r. \end{aligned}$$

In order to create a portfolio that is risk free, we need to set the coefficient of $dB_r$ equal to 0:

$$\begin{aligned} 0 &= \sigma_p(r,\tau_1)\,P(r,\tau_1) - x\sigma_p(r,\tau_2)\,P(r,\tau_2) \\ x\sigma_p(r,\tau_2)\,P(r,\tau_2) &= \sigma_p(r,\tau_1)\,P(r,\tau_1) \\ x &= \frac{\sigma_p(r,\tau_1)\,P(r,\tau_1)}{\sigma_p(r,\tau_2)\,P(r,\tau_2)}. \end{aligned}$$

**1.4** *Use the expression for the rate of return on portfolio $H(t)$ to derive an expression relating the Sharpe ratio of the first bond with maturity $\tau_1$ to the Sharpe ratio of the second bond with maturity $\tau_2$. Provide an economic interpretation for this condition.*

**Solution:**

Because portfolio $H(t)$ has no risk, its expected return must be equal to the risk-free rate of return:

$$\begin{aligned} r(t)H(t) &= \mu_p(r,\tau_1)P(r,\tau_1) - x\mu_p(r,\tau_2)P(r,\tau_2) \\ r(t)\Big[1\,P(r,\tau_1) - x\,P(r,\tau_2)\Big] &= \mu_p(r,\tau_1)P(r,\tau_1) - x\mu_p(r,\tau_2)P(r,\tau_2) \\ r(t)P(r,\tau_1) - x\,r(t)P(r,\tau_2) &= \mu_p(r,\tau_1)P(r,\tau_1) - x\mu_p(r,\tau_2)P(r,\tau_2) \end{aligned}$$

which, upon collecting terms for the bonds maturating at $\tau_1$ and $\tau_2$, leads to:

$$\begin{aligned} \mu_p(r,\tau_1)P(r,\tau_1) - r(t)P(r,\tau_1) &= x\mu_p(r,\tau_2)P(r,\tau_2) - x\,r(t)P(r,\tau_2) \\ P(r,\tau_1)\Big[\mu_p(r,\tau_1) - r(t)\Big] &= xP(r,\tau_2)\Big[\mu_p(r,\tau_2) - r(t)\Big] \end{aligned}$$

and now we substitute in the expression for $x$:

$$P(r,\tau_1)\Big[\mu_p(r,\tau_1) - r(t)\Big] = \left(\frac{\sigma_p(r,\tau_1)\,P(r,\tau_1)}{\sigma_p(r,\tau_2)\,P(r,\tau_2)}\right) P(r,\tau_2)\Big[\mu_p(r,\tau_2) - r(t)\Big],$$

which, on simplifying, leads to:

$$\frac{\mu_p(r,\tau_1) - r(t)}{\sigma_p(r,\tau_1)} = \frac{\mu_p(r,\tau_2) - r(t)}{\sigma_p(r,\tau_2)};$$

that is, bonds of different maturities have the *same* Sharpe ratio, implying that all bonds offer the same (excess) return-to-risk tradeoff.

**Q2. Portfolio choice using dynamic programming**

We consider a securities market model with two securities. The first security is a bond, with price process

$$\frac{dR(t)}{R(t)} = r\,dt,$$

where $r$ is a constant. The second security is a stock, with price process

$$\frac{dS(t)}{S(t)} = \mu(t)dt + \sigma\,dB(t),$$

where $\sigma$ is a constant, and $\mu(t)$ is given by

$$d\mu(t) = \kappa\big(\theta - \mu(t)\big)dt - s\,dB(t),$$

for some constants $\kappa$, $\theta$, and $s$; that is, the expected return on the stock is stochastic.

An investor with initial wealth $W$, maximizes utility over terminal consumption, where utility has constant relative risk aversion:

$$U\big(C(T)\big) = \frac{\big[C(T)\big]^{1-\gamma}}{1-\gamma}.$$

Based on this information, please answer the following questions.

**2.1** Define $\eta(t)$ as

$$\eta(t) \equiv \frac{\mu(t) - r}{\sigma},$$

and $\theta_\eta \equiv (\theta - r)/\sigma$ and $s_\eta \equiv s/\sigma$. Determine the process followed by $\eta(t)$.

**2.2** Formulate the investor's problem as a dynamic programming problem, denoting by $w$ the fraction of wealth invested in the risky asset at time $t$. That is, write the objective function and the dynamic budget constraint.

**2.3** Explain why the investor's value function can be written in the form $J\big(W(t), \eta(t), t\big)$; that is, why is $J$ a function of $W(t)$, $\eta(t)$, and $t$. Then, write down the Bellman equation that characterizes $J\big(W(t), \eta(t), t\big)$.

**2.4** From the Bellman equation, find the expression for the optimal proportion of wealth invested in the risky asset, $w(t)$, and provide an interpretation for it.

**2.5** Explain whether or not this problem can be solved using the martingale pricing approach, where the budget constraint is written in a static rather than dynamic form.

**2.1** *Define $\eta(t)$ as*

$$\eta(t) \equiv \frac{\mu(t) - r}{\sigma},$$

*and determine the process followed by $\eta(t)$. Define $\theta_\eta \equiv (\theta - r)/\sigma$ and $s_\eta \equiv s/\sigma$.*

**Solution:**
We have

$$\begin{aligned} d\eta_t &= \frac{d\mu_t}{\sigma} \\ &= \frac{\kappa(\theta - \mu_t)}{\sigma}dt - \frac{s}{\sigma}dB_t \\ &= \frac{\kappa[\theta - (r + \sigma\eta_t)]}{\sigma}dt - \frac{s}{\sigma}dB_t \\ &= \kappa(\theta_\eta - \eta_t)dt - s_\eta dB_t. \end{aligned}$$

**2.2** *Formulate the investor's problem as a dynamic programming problem, denoting by $w$ the fraction of wealth invested in the risky asset at time $t$. That is, write the objective function and the dynamic budget constraint.*

**Solution:**
The investor's problem is to choose a control $w_t$ to maximize

$$\max_{w_t} E\left[\frac{C_T^{1-\gamma}}{1-\gamma}\right],$$

subject to

$$dW_t = W_t\left[r + w_t(\mu_t - r)\right]dt + W_t w_t \sigma dB_t.$$

**2.3** *Explain why the investor's value function can be written in the form $J\big(W(t), \eta(t), t\big)$; that is, why is $J$ a function of $W(t)$, $\eta(t)$, and $t$. Then, write down the Bellman equation that characterizes $J\big(W(t), \eta(t), t\big)$.*

**Solution:**
The value function at time $t$ depends on $W_t$, $t$, and on state variables that affect the investment opportunity set. The only such state variable is $\eta_t$ (or equivalently $\mu_t$).
The Bellman equation is

$$\max_{w_t}\left[D_{W\eta}V + V_t\right] = 0,$$

where

$$\begin{aligned} D_{W\eta}V &= V_W W_t\left[r + w_t(\mu_t - r)\right] + V_\eta\kappa(\theta_\eta - \eta_t) \\ &\quad + \frac{1}{2}\left[V_{WW}W_t^2 w_t^2\sigma^2 - 2V_{W\eta}W_t w_t\sigma s_\eta + V_{\eta\eta}s_\eta^2\right]. \end{aligned}$$

**2.4** *From the Bellman equation, find the expression for the optimal proportion of wealth invested in the risky asset, $w(t)$, and provide an interpretation for $w(t)$.*

**Solution:**

The first-order condition w.r.t. $w_t$ is

$$w_t = -\frac{V_W}{W_t V_{WW}} \frac{\mu_t - r}{\sigma^2} + \frac{V_{W\eta}}{W_t V_{WW}} \frac{s_\eta}{\sigma}, \tag{1}$$

where the first term on the right-hand side is the myopic portfolio and the second term is the intertemporal hedging portfolio.

**2.5** *Explain whether or not this problem can be solved using the martingale pricing approach, where the budget constraint is written in a static rather than dynamic form.*

**Solution:**

The answer to this question depends on whether markets are complete or incomplete.

- If markets are complete, then it is straightforward to use the static martingale approach to solve this problem. In the setting described in the question, markets **are** complete – there is only a single Brownian driving all shocks and this is "spanned" by the risk-free bond and stock. Thus, it is possible to solve the problem using the standard martingale pricing approach.
- If markets were incomplete, then one could not use the static martingale approach as we had discussed in class, because to use that approach one needs to write an expression that says that the present value of future consumption is equal to the current (initial) value of wealth.
- When markets are not complete, we do not have a full set of Arrow-Debreu state prices, and so it is not straightforward to find the current value of future consumption (because we do know how exactly to discount future consumption, which is random, to obtain its present value).
- However, it is possible to extend the static martingale approach we developed in class in order to apply it in a setting where markets are incomplete.

**Q3. Intertemporal consumption and portfolio choice with two risky assets**

We consider the intertemporal consumption-portfolio problem of an investor "$a$". The objective of this investor is to choose her portfolio policy and consumption policy to maximize a time-separable expected utility function that depends on the rate of consumption at all future dates:

$$\max E_0 \left[ \int_0^\infty e^{-\delta t} \frac{C^{1-\gamma_a}}{1-\gamma_a} \, ds \right].$$

Suppose that the investor can allocate wealth across 2 risky assets and a bank account. The price $S_n$ of each of the risky assets is given by:

$$\frac{dS_n}{S_n} = \mu_n \, dt + \sigma_n \, dB_n, \qquad n = \{1, 2\},$$

where $\mu_n$ and $\sigma_n$ are constants, $B_n$ is a Brownian motion (under the physical measure), and the correlation between $B_1$ and $B_2$ is zero. Let the risk-free interest rate earned on money invested in the bank account be denoted by $r$, which is constant.

Denote the value function of the investor by $J$ and the proportion of wealth allocated to the two risky assets by $w_1$ and $w_2$.

Based on the above, please answer the following three questions.

**3.1** Identify the state variables of the investor's problem and write the dynamic optimization problem of the investor along with his budget constraint.

**3.2** What is the the Bellman equation that characterizes the solution to the investor's problem?

**3.3** Derive the first-order conditions for the investor's portfolio problem.

**3.4** Show that the investors's optimal portfolio policy can be written as an investment in two funds: (i) a fund consisting of *only* the risk-free asset; (ii) a fund consisting of *only* the two risky assets (please identify the weights assigned to each of the two risky assets in this fund; denote these weights by $x_1$ and $x_2$).

**3.5** Now consider the intertemporal consumption-portfolio problem of investor "$b$" who wishes to maximize:

$$\max E_0 \left[ \int_0^\infty e^{-\delta t} \frac{C^{1-\gamma_b}}{1-\gamma_b} \, ds \right].$$

Will this investor choose to invest in the same two funds as investor "$a$" or will she want to invest in funds that are different? Explain why.

**3.1** *Identify the state variables of the investor's problem and write the dynamic optimization problem of the investor along with his budget constraint.*

**Solution:**

Because the investment opportunity set is constant, and the horizon is infinite, the only state variable for this problem is the investor's wealth.

Therefore, the investor's optimization problem can be expressed as:

$$J(W) = \max_{C,w} E_0 \left[ \int_0^\infty e^{-\delta t} \frac{C^{1-\gamma_a}}{1-\gamma_a} \, ds \right],$$

subject to the dynamic budget constraint:

$$\frac{dW}{W} = \Big[ w_1\mu_1 + w_2\mu_2 + \big(1 - w_1 - w_2\big)r - \frac{C}{W} \Big] dt + w_1\sigma_1 dB_1 + w_2\sigma_2 dB_2.$$

**3.2** *What is the Bellman equation that characterizes the solution to the investor's problem.*

**Solution:**

The Bellman equation is:

$$\begin{aligned} 0 = e^{-\delta t} \frac{C^{1-\gamma_a}}{1-\gamma_a} + J_t + W J_W \Big[ w_1\mu_1 + w_2\mu_2 + \big(1 - w_1 - w_2\big)r - \frac{C}{W} \Big] \\ + \frac{1}{2} W^2 J_{WW} \Big( w_1^2\sigma_1^2 + w_2^2\sigma_2^2 \Big). \end{aligned}$$

**3.3** *Derive the first-order conditions for the investor's portfolio problem.*

**Solution:**

The first-order conditions for the portfolio policies are:

$$\begin{aligned} 0 &= W J_W (\mu_n - r) + W^2 J_{WW} w_n \sigma_n^2, \qquad n = \{1, 2\} \\ w_n &= -\frac{J_W}{W J_{WW}} \frac{(\mu_n - r)}{\sigma_n^2}, \qquad n = \{1, 2\}, \\ w_n &= \frac{1}{\gamma_a} \frac{(\mu_n - r)}{\sigma_n^2}, \qquad n = \{1, 2\} \end{aligned}$$

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**3.4** *Show that the investors's optimal portfolio policy can be written as an investment in two funds: (i) a fund consisting of just the risk-free asset; (ii) a fund consisting of just the two risky assets (please identify the weights assigned to each of the two risky assets in this fund).*

**Solution:**

Above, we have derived the expressions for the optimal proportion of wealth to be

invested in the two assets, $w_1$ and $w_2$. The proportion of wealth to be invested in the risk-free asset is then given by $1 - w_1 - w_2$.

Now, define the two funds to be: (i) a fund with just the risk-free asset; and, (ii) a fund with the just the two risky assets in the following proportion:

$$x_n = \frac{w_n}{w_1 + w_2} = \frac{\frac{1}{\gamma_a}\frac{(\mu_n - r)}{\sigma_n^2}}{\frac{1}{\gamma_a}\frac{(\mu_1 - r)}{\sigma_1^2} + \frac{1}{\gamma_a}\frac{(\mu_2 - r)}{\sigma_2^2}} = \frac{\frac{(\mu_n - r)}{\sigma_n^2}}{\frac{(\mu_1 - r)}{\sigma_1^2} + \frac{(\mu_2 - r)}{\sigma_2^2}},$$

which we observe is independent of the investor's relative risk aversion $\gamma_a$; (or, equivalently, is independent of $\frac{-J_W}{W J_{WW}}$). This result is referred to as "two-fund separation."

**3.5** *Now consider the intertemporal consumption-portfolio problem of investor "b" who wishes to maximize:*

$$\max E_0 \left[ \int_0^\infty e^{-\delta t} \frac{C^{1-\gamma_b}}{1-\gamma_b} \, ds \right].$$

*Will this investor choose to invest in the same two funds as investor "a" or will she want to invest in funds that are different? Explain why.*

**Solution:**

Yes, this investor will also choose to invest in the same two funds, but in different proportions depending on how $\gamma_b$ compares to $\gamma_a$. For instance, if $\gamma_b < \gamma_a$, then investor $b$ is less risk averse than investor $a$, so investor $b$ will invest less in the fund with the risk-free asset and more in the fund with just the two risky assets.

**Q4. Intertemporal consumption, portfolio, and labor choice**

Consider an individual's intertemporal consumption, *labor*, and asset-allocation problem for the case of a risk-free asset and a single risky asset.

The individual maximizes her expected lifetime utility, which is of the form:

$$E_0\left[\int_0^T e^{-\delta t}u(C_t, L_t)dt\right],$$

where $C_t$ is the individual's consumption at date $t$ and $L_t$ is the amount of effort that the individual exerts at date $t$. The function $u(C_t, L_t)$ is assumed to be an increasing concave function of $C_t$ but a decreasing concave function of $L_t$.

The risk-free asset pays a constant rate of return equal to $r$ per unit of time. The price of the risk asset $S(t)$ satisfies the process:

$$\frac{dS(t)}{S(t)} = \mu_s\, dt + \sigma_s\, dB_s,$$

where $\mu_s$ and $\sigma_s$ are constants.

For each unit of labor effort exerted at date $t$, the individual receives an instantaneous flow of labor income of $L(t)\, y(t)\, dt$. The return to effort, or the wage rate, $y(t)$, is stochastic and follows the process:

$$dy(t) = \mu_y\, dt + \sigma_y\, dB_y,$$

where $\mu_y$ and $\sigma_y$ are constants and the correlation between the Brownians is: $(dB_s) \times (dB_y) = \rho\, dt$.

Based on the above, please answer the following questions.

**4.1** Letting $W$ denote the investor's wealth and $w$ the proportion of wealth invested in the risky asset, write down the dynamic (intertemporal) budget constraint for this problem.

**4.2** Denote the value function by $J$. What are the state variables for the optimization problem of the investor? Write down the Bellman equation?

**4.3** Write the first-order conditions that the optimal consumption, labor-effort, and portfolio policies must satisfy.

**4.4** Show how the optimal demand for the risky asset, $w$, can be written as the sum of two terms: one term that would be present even if $y$ was constant and another term that is present only because $y$ is stochastic. Provide an economic interpretation for the two terms.

**4.1** *Letting $W$ denote the investor's wealth and $w$ the proportion of wealth invested in the risky asset, write down the dynamic (intertemporal) budget constraint for this problem.*

**Solution:**
The dynamic budget equation is:

$$\begin{aligned} dW &= W\left[w\frac{dS}{S} + (1-w)r\,dt\right] - C\,dt + L\,y\,dt \\ &= W\left[w\Big(\mu_s\,dt + \sigma_s\,dB_s\Big) + (1-w)r\,dt\right] - C\,dt + L\,y\,dt \\ &= \Big[wW\Big(\mu_s - r\Big) + r\,W - C + L\,y\Big]\,dt + W\,w\,\sigma_s\,dB_s. \end{aligned}$$

**4.2** *Denote the value function by $J$. What are the state variables for the optimization problem of the investor? Write down the Bellman equation?*

**Solution:**
The state variables for the investor's problem are time, the investor's wealth, and the wage rate, which is denoted by $y$. Thus, the value function can be written as $J(t, W, y)$.
The Bellman equation is given by the following:

$$\begin{aligned} 0 &= \max_{C,L,w}\left\{e^{-\delta t}u(C_t, L_t) + \frac{1}{dt}E\Big[dJ\Big]\right\} \\ &= \max_{C,L,w}\Bigg\{e^{-\delta t}u(C_t, L_t) + J_t \\ &\qquad + J_W\left[wW\Big(\mu_s - r\Big) + r\,W - C + L\,y\,\right] + \frac{1}{2}J_{WW}\left[W^2\,w^2\,\sigma_s^2\right] \\ &\qquad + J_y\mu_y + \frac{1}{2}J_{yy}\sigma_y^2 + J_{Wy}W\,w\,\sigma_s\sigma_y\rho\Bigg\}. \end{aligned}$$

**4.3** *Write the first-order conditions that the optimal consumption, labor-effort, and portfolio policies must satisfy.*

**Solution:**
The first-order conditions with respect to the three choice variables are:

$$\begin{aligned} \frac{\partial}{\partial C}: &\qquad 0 = e^{-\delta t}\frac{\partial u(C_t, L_t)}{\partial C} - J_W; \\ \frac{\partial}{\partial L}: &\qquad 0 = e^{-\delta t}\frac{\partial u(C_t, L_t)}{\partial L} + y\,J_W; \\ \frac{\partial}{\partial w}: &\qquad 0 = J_W W\Big(\mu_s - r\Big) + J_{WW}\left[W^2\,w\,\sigma_s^2\right] + J_{Wy}W\,\sigma_s\sigma_y\rho. \end{aligned}$$

**4.4** *Show how the demand for the risky asset, $w$, can be written as the sum of two terms: one term that would be present even if $y$ was constant and another term that is present only because $y$ is stochastic. Provide an economic interpretation for the two terms.*

**Solution:**
From the first-order condition for the optimal portfolio weight (the last equation above), solving for $w$ gives:

$$\begin{aligned} w &= -\frac{J_W W\left(\mu_s - r\right)}{J_{WW}\left[W^2\sigma_s^2\right]} - \frac{J_{Wy} W\,\sigma_s\sigma_y\rho}{J_{WW}\left[W^2\sigma_s^2\right]} \\ &= -\frac{J_W}{W\,J_{WW}}\frac{\left(\mu_s - r\right)}{\sigma_s^2} - \frac{J_{Wy}}{W J_{WW}}\frac{\sigma_s\sigma_y\rho}{\sigma_s^2} \\ &= -\frac{J_W}{W\,J_{WW}}\frac{\left(\mu_s - r\right)}{\sigma_s^2} - \frac{J_{Wy}}{W J_{WW}}\frac{\mathsf{cov}(S,y)}{\mathsf{var}(S)} \\ &= -\frac{J_W}{W\,J_{WW}}\frac{\left(\mu_s - r\right)}{\sigma_s^2} - \frac{J_W}{W J_{WW}}\frac{J_{Wy}}{J_W}\mathsf{beta}(S,y), \end{aligned}$$

where one can interpret

- the first component as the "myopic" portfolio of a mean-variance investor, and
- the second component as the "intertemporal hedge" against changes in the wage rate, $y$.
- The term $-J_W/(WJ_{WW})$ is the risk tolerance of the investor (the reciprocal of risk aversion), and the investment in the risky asset increases with risk tolerance.
- The term $J_{Wy}/J_W$ is the elasticity of marginal utility with respect to the wage rate, $y$; that is, it is $\partial \ln J_W/\partial y$.

**Q5. Portfolio choice with inflation**

Consider the following consumption and portfolio choice problem. An individual must choose between two different assets, a stock and a short (instantaneous) maturity, default-free bond. *In addition, the individual faces a stochastic rate of inflation*, that is, uncertain changes in the price level (e.g., the Consumer Price Index) denoted by $P(t)$.

The *price level* (currency price of the consumption good) follows the process

$$\frac{dP(t)}{P(t)} = \mu_\pi \, dt + \sigma_\pi \, dB_\pi(t).$$

The nominal (currency value) of the stock is $S(t)$ and it satisfies the process

$$\frac{dS(t)}{S(t)} = \mu_S \, dt + \sigma_S \, dB_S(t).$$

The nominal (currency value) of the bond (bank deposit) is given by $D(t)$ . It pays an instantaneous nominal rate of return equal to $r$. Hence, its nominal price satisfies the process

$$\frac{dD(t)}{D(t)} = r \, dt.$$

Assume that $\mu_\pi$, $\sigma_\pi$, $\mu_S$, $\sigma_S$, and $r$ are constant. Assume also that the correlation $\rho$ between the two Brownian motions is constant: $dB_\pi(t) \times dB_S(t) = \rho \, dt$.

Based on the above, please answer the following questions.

**5.1** Denote the real *price* of the stock by $R^S(t) = S(t)/P(t)$, and denote the real *price* of the bank deposit by $R^D(t) = D(t)/P(t)$. What processes do the real (consumption-good value) rates of return on the stock and the bond satisfy; that is, find $dR^S(t)/R^S(t)$ and $dR^D(t)/R^D(t)$?

**5.2** Let $c(t)$ be the individual's date $t$ *real* rate of consumption and let $w(t)$ be the proportion of real wealth, $W(t)$, that is invested in the stock. Derive the process followed by real wealth. That is, find the dynamic budget constraint.

**5.3** Assume that an infinitely-lived individual chooses $c(t)$ and $w(t)$ to solve the following problem:

$$\max_{c(t),w(t)} E_0 \int_0^\infty U\big[c(t), t\big] dt,$$

subject to the dynamic budget constraint derived in Q 5.2 above. Assuming that $U\big[c(t), t\big]$ is a concave utility function, solve for the individual's optimal choice of $w(t)$ in terms of the indirect utility-of-wealth function, $J$.

**5.4** How does the optimal portfolio weight $w(t)$ vary with the correlation $\rho$? What is the economic intuition for this comparative-static result?

**5.1** *Denote the real* price *of the stock by* $R^S(t) = S(t)/P(t)$*, and denote the real* price *of the bank deposit by* $R^D(t) = D(t)/P(t)$*. What processes do the real (consumption-good value) rates of return on the stock and the bond satisfy; that is, find* $dR^S(t)/R^S(t)$ *and* $dR^D(t)/R^D(t)$*?*

**Solution:**

To find the real return on the stock and the bond, we will use Itô's formula. As a preliminary step, we find the first, second, and cross-derivates of $R^S(t)$ and $R^D(t)$.

$$\frac{\partial R^S(t)}{\partial S(t)} = \frac{1}{P(t)}; \qquad \frac{\partial R^D(t)}{\partial D(t)} = \frac{1}{P(t)};$$

$$\frac{\partial^2 R^S(t)}{\partial S^2(t)} = 0; \qquad \frac{\partial^2 R^D(t)}{\partial D^2(t)} = 0;$$

$$\frac{\partial R^S(t)}{\partial P(t)} = \frac{-S(t)}{P^2(t)}; \qquad \frac{\partial R^D(t)}{\partial P(t)} = \frac{-D(t)}{P^2(t)};$$

$$\frac{\partial^2 R^S(t)}{\partial P^2(t)} = \frac{2S(t)}{P^3(t)}; \qquad \frac{\partial^2 R^D(t)}{\partial P^2(t)} = \frac{2D(t)}{P^3(t)};$$

$$\frac{\partial^2 R^S(t)}{\partial S(t)\partial P(t)} = \frac{-1}{P^2(t)}; \qquad \frac{\partial^2 R^D(t)}{\partial D(t)\partial P(t)} = \frac{-1}{P^2(t)};$$

Then,

$$dR^S(t) = \frac{\partial R^S(t)}{\partial S(t)}dS(t) + \frac{1}{2}\frac{\partial^2 R^S(t)}{\partial S^2(t)}(dS(t))^2 + \frac{\partial R^S(t)}{\partial P(t)}dP(t) + \frac{1}{2}\frac{\partial^2 R^S(t)}{\partial P^2(t)}(dP(t))^2 + \frac{\partial^2 R^S(t)}{\partial S(t)\partial P(t)}dS(t)dP(t)$$

Substituting the values of the partial derivatives given above, we get:

$$dR^S(t) = \frac{1}{P(t)}dS(t) + \frac{1}{2}0 - \frac{S(t)}{P^2(t)}dP(t) + \frac{1}{2}\frac{2S(t)}{P^3(t)}(dP(t))^2 - \frac{1}{P^2(t)}dS(t)dP(t)$$

which implies that the real *return* on the stock is

$$\begin{aligned}\frac{dR^S(t)}{R^S(t)} &= (\mu_S dt + \sigma_S dB_S) - (\mu_\pi dt + \sigma_\pi dB_\pi) + \sigma_\pi^2 dt - \rho\sigma_S\sigma_\pi dt \\ &= \Big(\mu_S - \mu_\pi + \sigma_\pi^2 - \rho\sigma_S\sigma_\pi\Big)dt + \sigma_S dB_S - \sigma_\pi dB_\pi.\end{aligned}$$

Similarly, the real *return* on the bank deposit is

$$dR^D(t) = \frac{1}{P(t)}dD(t) + 0 - \frac{D(t)}{P^2(t)}dP(t) + \frac{1}{2}\frac{2D(t)}{P^3(t)}(dP(t))^2 - \frac{1}{P^2(t)}dD(t)dP(t)$$

$$\begin{aligned}\frac{dR^D(t)}{R^D(t)} &= r\,dt - (\mu_\pi dt + \sigma_\pi dB_\pi) + \sigma_\pi^2 dt \\ &= \Big(r - \mu_\pi + \sigma_\pi^2\Big)dt - \sigma_\pi dB_\pi.\end{aligned}$$

**5.2** *Let $c(t)$ be the individual's date $t$* real *rate of consumption and let $w(t)$ be the proportion of real wealth, $W(t)$, that is invested in the stock. Derive the process followed by real wealth.*

**Solution:**

The dynamic budget constraint is given by:

$$\begin{aligned}
dW(t) &= w(t)W(t)\Big[\Big(\mu_S - \mu_\pi + \sigma_\pi^2 - \rho\sigma_S\sigma_\pi\Big)dt + \sigma_S dB_S - \sigma_\pi dB_\pi\Big] \\
&\quad + (1 - w(t))W(t)\Big[\Big(r - \mu_\pi + \sigma_\pi^2\Big)dt - \sigma_\pi dB_\pi\Big] - c(t)\,dt. \\
&= w(t)W(t)\Big[\Big(\mu_S - \mu_\pi + \sigma_\pi^2 - \rho\sigma_S\sigma_\pi\Big) - \Big(r - \mu_\pi + \sigma_\pi^2\Big)\Big]dt \\
&\quad + w(t)W(t)\Big[\Big(\sigma_S dB_S - \sigma_\pi dB_\pi\Big) + \sigma_\pi dB_\pi\Big] \\
&\quad + W(t)\Big[\Big(r - \mu_\pi + \sigma_\pi^2\Big)dt - \sigma_\pi dB_\pi\Big] - c(t)\,dt \\
&= \Big[w(t)W(t)\Big(\mu_S - r - \rho\sigma_S\sigma_\pi\Big) + W(t)\Big(r - \mu_\pi + \sigma_\pi^2\Big) - c(t)\Big]dt \\
&\quad + w(t)W(t)\sigma_S dB_S - W(t)\sigma_\pi dB_\pi.
\end{aligned}$$

**5.3** *Assume that an infinitely-lived individual chooses $c(t)$ and $w(t)$ to solve the following problem:*

$$\max_{c(t),w(t)} E_0 \int_0^\infty U\big[c(t),t\big]dt,$$

*subject to the dynamic budget constraint derived in Q 2.2 above. Assuming that $U\big[c(t),t\big]$ is a concave utility function, solve for the individual's optimal choice of $w(t)$ in terms of the indirect utility-of-wealth function, $J$.*

**Solution:**

The investment opportunity set is constant in this question because all the coefficients of the price processes for the stock and the bank deposit and also for the inflation process are constant. Thus, the only state variable is wealth, $W(t)$. Thus, the Bellman equation for $J(W(t))$ is given by:

$$\begin{aligned}
0 &= \max_{c(t),w(t)} \Big\{U\big[c(t),t\big] + J_t + J_W E\big[dW\big] + \frac{1}{2}J_{WW}E\big[(dW)^2\big]\Big\} \\
&= \max_{c(t),w(t)} \Big\{U\big[c(t),t\big] + J_t + W(t)J_W\Big[w(t)\Big(\mu_S - r - \rho\sigma_S\sigma_\pi\Big) + \Big(r - \mu_\pi + \sigma_\pi^2\Big) - \frac{c(t)}{W(t)}\Big] \\
&\quad + \frac{1}{2}W^2(t)J_{WW}\Big[w^2(t)\sigma_S^2 + \sigma_\pi^2 - 2w(t)\rho\sigma_S\sigma_\pi\Big]\Big\}
\end{aligned}$$

The first-order condition with respect to consumption, $c(t)$, is:

$$0 = U_c\big[c(t),t\big] - J_W. \tag{2}$$

The first-order condition with respect to the portfolio weight, $w(t)$, is:

$$\begin{aligned}
0 &= W(t)J_W\Big(\mu_S - r - \rho\sigma_S\sigma_\pi\Big) + W^2(t)J_{WW}\Big[w(t)\sigma_S^2 - \rho\sigma_S\sigma_\pi\Big] \\
w(t) &= \frac{-J_W}{W(t)J_{WW}}\frac{(\mu_S - r - \rho\sigma_S\sigma_\pi)}{\sigma_S^2} + \frac{\rho\sigma_\pi}{\sigma_S}.
\end{aligned} \tag{3}$$

**5.4** *How does $w(t)$ vary with the correlation $\rho$? What is the economic intuition for this comparative-static result?*

**Solution:**

Differentiating the optimal $w(t)$ with respect to $\rho$, we get that:

$$\begin{aligned}
\frac{\partial w(t)}{\partial \rho} &= \frac{-J_W}{W(t)J_{WW}} \frac{(-\sigma_S \sigma_\pi)}{\sigma_S^2} + \frac{\sigma_\pi}{\sigma_S} \\
&= \frac{J_W}{W(t)J_{WW}} \frac{\sigma_\pi}{\sigma_S} + \frac{\sigma_\pi}{\sigma_S} \\
&= \left(1 - \frac{-J_W}{W(t)J_{WW}}\right) \frac{\sigma_\pi}{\sigma_S} \\
&= \left(1 - \frac{1}{\gamma}\right) \frac{\sigma_\pi}{\sigma_S}, \qquad (4)
\end{aligned}$$

where, the relative risk aversion of the investor, $\gamma$, is given by

$$\frac{1}{\gamma} = -\frac{-J_W}{W(t)J_{WW}}.$$

The above expression tells us how the investor uses the stock to hedge against inflation.

- If $\rho = 0$, then there is no correlation between stock returns and prices, so stocks are not used to hedge inflation at all.
- If $\rho \neq 0$, then stocks are used to hedge inflation.
- The size of the hedge increases in magnitude as the relative risk aversion of the investor deviates from 1. This is because when the relative risk aversion of the investor is 1, then the income and elasticity effects exactly offset one another, and the hedging demand is zero.
- The sign of the hedge depends on whether $\gamma$ is smaller or bigger than 1, which determines whether the substitution or income effect dominates.